

TapTen — a 10-second finger-tapping test, measured privately by any webcam

UnivaBio 2026 · AI for Human Health · individual entry · live: ryugi62.github.io/tapten · code: github.com/Ryugi62/tapten

Problem

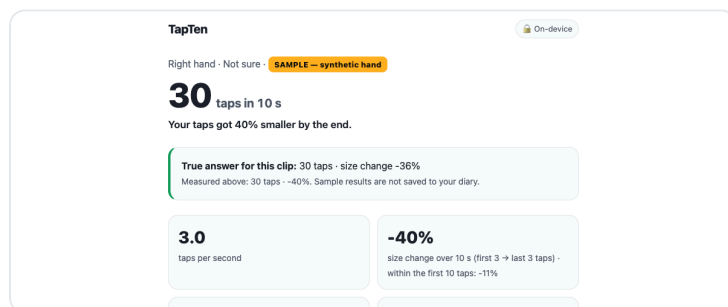
People with Parkinson's disease see a neurologist every few months. Between visits, how their movement changed when medication wore off is reported **from memory**. At the bedside, clinicians rate finger tapping (MDS-UPDRS item 3.4: 10 taps of index finger on thumb, fast and big) and watch for taps that get **slower and smaller** (the "decrement"). TapTen uses a fixed 10-second window so tests are comparable. Phone screen-tap tests capture speed but **cannot see how wide the fingers open**.

What TapTen does

- **Webcam test**: pick the hand, tag how you move right now (on / off, adapted from the Hauser diary) and time since your last levodopa dose; it starts by itself when the hand is visible and still, with beeps.
- **Numbers first**: taps in 10 s, taps/s, **size change from the first 3 to the last 3 taps**, speed change, pauses — a bar for every tap. One test is described in plain words (95 % of single tests are within about ± 19 pts on our benchmark); after 3 tests it is compared with your own usual range.
- **Quality gate**: refuses to report when the hand was found in < 85 % of frames, the camera was too slow (<15 fps), the hand was too small, or the test was too short.
- **Private diary** (export/import) and a **one-page clinic sheet** with medians per state and taps vs minutes since dose. Never says what to do with medication.

How it works

AI part: Google MediaPipe Hand Landmarker (pre-trained, Apache-2.0) finds 21 3D hand points per frame inside the browser — we trained no model. **Our part**: opening = thumb-tip ↔ index-tip distance ÷ palm length → robust tap detection → speed, size change, pauses; a quality gate; and a benchmark with exact answers.



Is it accurate? A ground-truth benchmark

No patient videos yet (that needs ethics approval). So a rigged 3D hand is animated with a **known** tap schedule — 0.7–5 taps/s, shrinking 0–90 %, slowing, pauses, tremor, several views and distances, blur/dim/noise/15 fps — and run through the **same model and analysis code**. Two in-distribution held-out sets (same generator); the second was pre-registered in its own commit before running.

23/24

clean: exact
tap count

21/24

held-out 1:
exact

16/20

held-out 2
(severe,
slowing):
exact

6/6

bad
recordings
refused

Exact count overall 69/80 (86.3 %; our own target was 90 % — not met). Degraded exact 9/12. Size change vs truth: bias 0.1 pts, 95 % limits ± 19.4 pts (single-test errors are wide, so single tests are not interpreted); severe-decrement error 9.3 pts; slowing flagged 6/6. Every clip, misses included: docs/bench-table.md.

Privacy, enforced in code

The page's Content-Security-Policy is `connect-src 'self'`; model and engine are bundled; no account. Do not change medication based on TapTen — normal ranges are not known. While testing we found the tracking runtime tries to send a usage log to `odml.pa.googleapis.com` — **the policy blocks it** (browser console, 2026-10-06).

Who it is for

People already living with Parkinson's and their care partners, to bring a measured home record — taken at the times that matter (morning "off", after a dose) — to appointments. Neurologists get a dated table, not a recollection.

Finished vs. planned (honest scope)

Finished: webcam test (auto-start, beeps), video-file mode, quality gate, motor diary + clinic sheet, benchmark with held-out and negative sets, CI.

Planned: measuring real people (healthy volunteers, test-retest), then patients with their neurologists; both-hand asymmetry.

Not a diagnosis, not an MDS-UPDRS score; a research prototype, not a cleared medical device. Synthetic-hand accuracy is not clinical accuracy. Closest prior work: PARK (Islam et al., npj Digit Med 2023) scores severity from webcam video for research; VisionMD (npj Parkinson's Dis 2025) analyses recorded videos locally for clinicians; FastEval (PMC10853559). TapTen's difference: nothing uploaded, no score, a live quality gate, and medication timing on a clinic sheet.

Built by Taegeol Kim (student, Changwon National University, Korea) with AI coding assistance; SPEC.md and 36 unit tests cover the domain logic. Licenses: MIT (TapTen) · Apache-2.0 (MediaPipe) · MIT (three.js, WebXR hand model — benchmark only).